

To: EGR Corporation
From: Guido Fernandez, Jack Kroothoep, Stephanie Lin, Shaine Wei
Subject: Design Challenge 3: Happy Meals
Date: 18 March 2026

Executive Summary

EGR Corporation has been contacted by Fluke Toy Corporation to design a McDonald's Happy Meal Toy to be implemented later in the year. The Happy Meal Toy must be creative, fun, low-cost, sustainable, and safe for children to use, meaning it must pass standardized choke test requirements. EGR Corporation has assigned the team to create a final Happy Meal Toy in time for the "Duke Toy Fair" event, where all design team proposals will be presented.

The team's ideation process involved rounds of brainstorming potential toy ideas through rapid generation of ideas without limits. Once a general idea was agreed upon, the team focused on refining said idea. The design and prototyping process involved iterative testing of individual CAD designs of the upper and lower bodies of the barbarian, the lever arm of the catapult, and the base of the catapult, all of which assemble to the complete Happy Meal Toy.

The team designed an interactive, medieval-theme Happy Meal Toy: Siege the Waste™, that includes a catapult and a barbarian. The user will use the food wrappers that come with the meal, crumbling them to create a paper projectile to load into the catapult to launch at the barbarian.

Approach Description

Ideation Techniques Used

The team began the design process using a brainstorming ideation method. During this session, team members proposed several toy concepts and sketched potential designs on a whiteboard. The brainstorming approach was chosen because it allows rapid generation of creative ideas without initially limiting design possibilities. Ideas included vehicles, a voice changer, character figures, mechanical toys, and launching mechanisms. Brainstorming encouraged collaboration and allowed each team member to contribute unique concepts while building on the ideas of others.

Idea Refinement and Selection

After generating several possible concepts, the team evaluated each idea based on several key criteria: safety, manufacturing simplicity, creativity, user interaction, and potential for incorporating sustainability. The catapult-style launching toy was selected because it provides interactive gameplay while maintaining a simple mechanical design that could be easily manufactured. Other ideas, such as the voice changer, failed to fit the budget or scope of the project. A medieval theme was chosen to enhance the play experience, aimed specifically towards a younger audience. This idea was further refined by identifying key components for the

toy: a catapult base, a catapult arm, and a two-piece break-apart target figure. The team also decided to add a sustainability feature in which children reuse packaging material as the projectile instead of including a plastic ball.

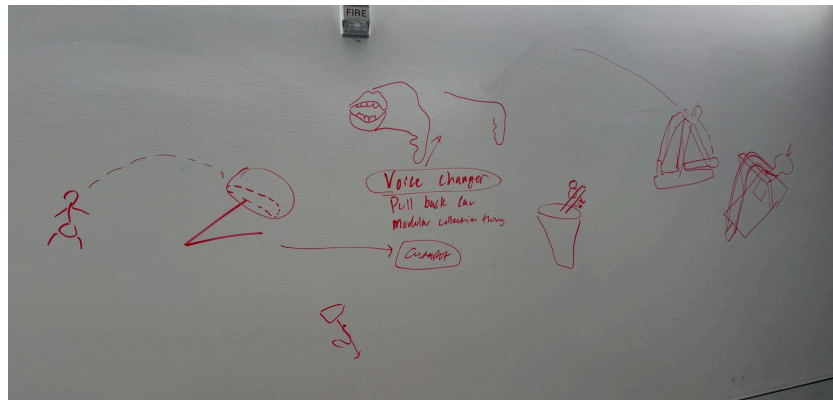


Figure 1. Visual of team brainstorming and sketching ideas

Toy Description and Documentation

Description

The team designed an interactive, medieval-theme Happy Meal Toy that includes a catapult (base and lever arm) and a barbarian with detachable lower and upper bodies. The user will use the food wrappers that come with the meal, crumbling them to create a paper projectile to load into the catapult. By quickly depressing the opposite end of the catapult with your finger, the projectile is launched at the barbarian. Upon falling over, the barbarian's torso detaches from the legs and the user is victorious.

Design Documentation for Torso

Since the torso is assembled using pivot pins through aligned holes, many dimensions are designated as critical-to-function (CTF) features to ensure proper alignment between the two torso halves and reliable operation of the arm socket interface. Although there is dimensional variation, both torso halves are produced from identical molds, resulting in consistent, symmetric tolerances between the mating parts. Since corresponding features vary in the same manner, relative alignment between the two halves is maintained, reducing the impact of tolerance stack-up, and thus does not require additional consideration as separate CTF stack-up conditions.

The critical-to-fit/function (CTF) dimensions across the Blank Side torso (Figure 2.1), Face Side torso (Figure 2.2), and Figure Arm (Figure 2.3) are those that directly govern assembly alignment, structural integrity, motion, and overall figure safety. The Ø0.18 pivot holes on Figure 2.2 and the Ø0.15 pivot pins, along with their locations relative to the top face, are primary CTF features because they must align precisely to allow proper pin insertion and secure assembly; any misalignment would prevent assembly or impair function. For the arm (Figure 2.3), additional CTF dimensions include the Ø0.30 socket head, the 0.12 offset between the main arm body and

socket head, and the Ø0.15 arm hole diameter, which all define the arm's geometry and range of motion.

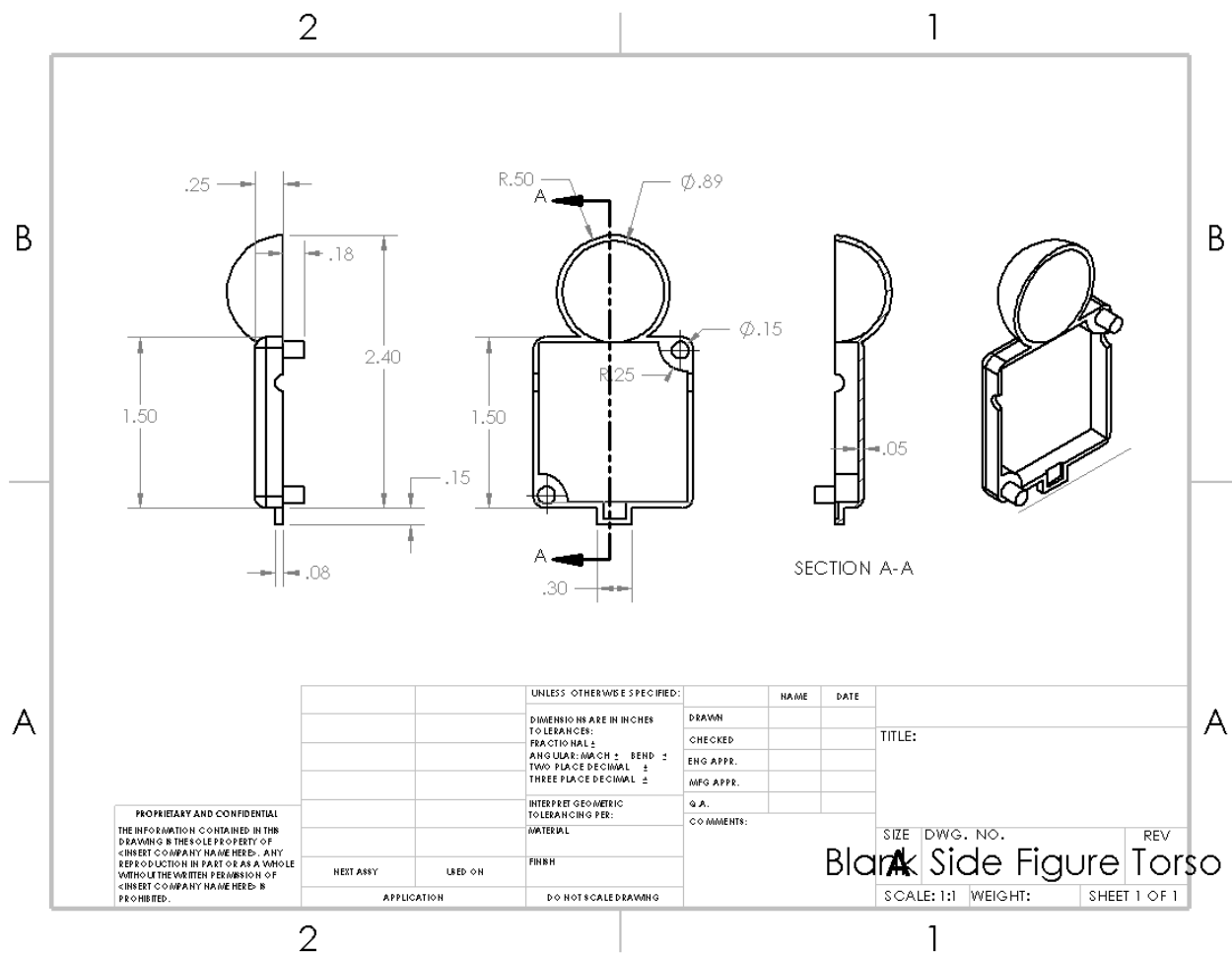


Figure 2.1. Blank Side Figure Torso Part Drawing

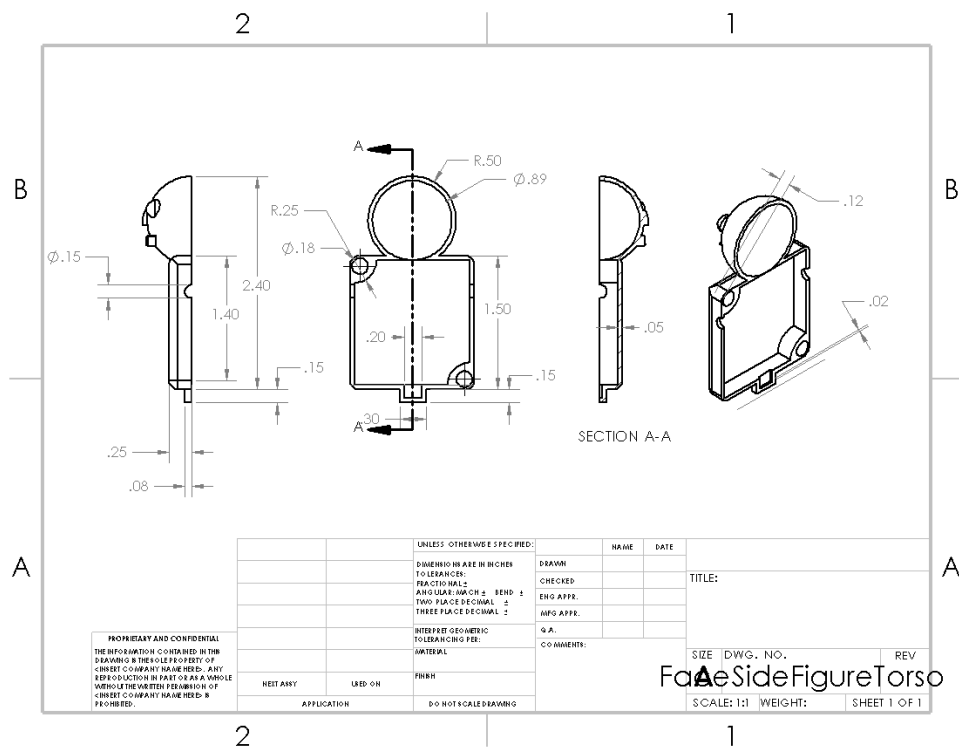


Figure 2.2. Face Side Figure Torso Part Drawing

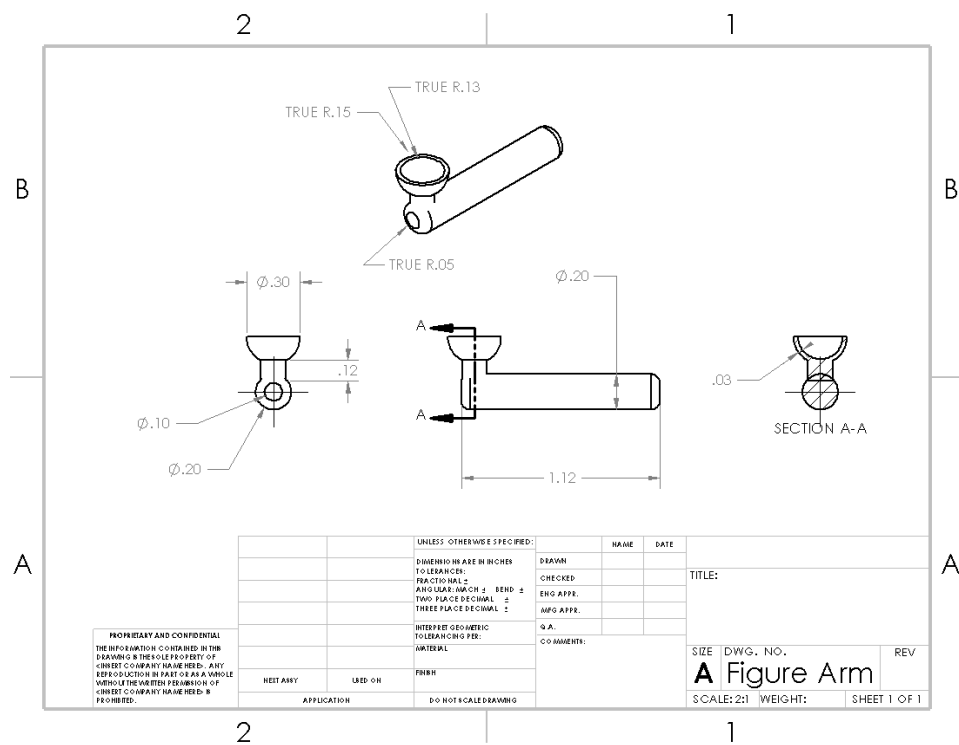


Figure 2.3. Figure Arm Part Drawing

Table 2.1 CTF Dimensions and Tolerances

Dimension Description	Nominal (in)	Tolerance ($\pm$ in)	Min (in)	Max (in)
Pivot Holes	$\varnothing$ 0.18	0.001	0.179	0.181
Pivot Pins	$\varnothing$ 0.15	0.002	0.148	0.152
Pivot Hole Support	R0.25	0.005	0.245	0.255
Socket Head	$\varnothing$ 0.30	0.002	0.298	0.302
Socket-Arm Offset	0.12	0.001	0.119	0.121
Arm Hole	$\varnothing$ 0.15	0.002	0.148	0.152
Slot Length	0.30	0.001	0.299	0.301
Slot Width	0.08	0.001	0.079	0.081
Slot Depth	0.15	0.001	0.149	0.151

Table 2.2 Pin Hole Interface Worst Case Clearance

Case	Calculation	Clearance (in)
Minimum Clearance	0.179-0.148	0.031
Maximum Clearance	0.181-0.152	0.029

Table 2.3 Socket Arm Interface Worst Case Clearance

Case	Calculation	Clearance (in)
Minimum Clearance	0.148-0.119	0.029
Maximum Clearance	0.152-0.121	0.031

Design Documentation for Legs

The CTF dimensions for the external envelope, the overall, outermost shape, ensure the proper fit with the back-leg part. It maintains safe clearances and structural integrity. The 1.93 inch overall height and 1.61-1.68 inch vertical locations of the key features are CTF because they match the back-leg and torso dimensions, guaranteeing that the assembled legs align and the feet make contact with the ground simultaneously, allowing the legs to stand. The 1.4 inch and 0.6 inch

length locations along the leg establish the main profile and are CTF since they control how the front leg part connects to the back leg part, avoiding gaps. The 0.55 inch and 0.44-0.25 inch widths across the mourning and lower regions are CTF since it determines clearance between the legs and its surrounding components. This prevents the parts interfering during any motion when in use. The 0.15 inch slot features with the 0.11-0.12 inch hole dimensions are CTF because they define the snap-fit design, allowing for the front and back leg parts to be connected. Improper tolerance could cause loose joints, making assembly between the two halves impossible. The 0.10-0.18 inch wall thickness from shelling the inside of the legs allows for manufacturability, providing enough material to withstand playtime.

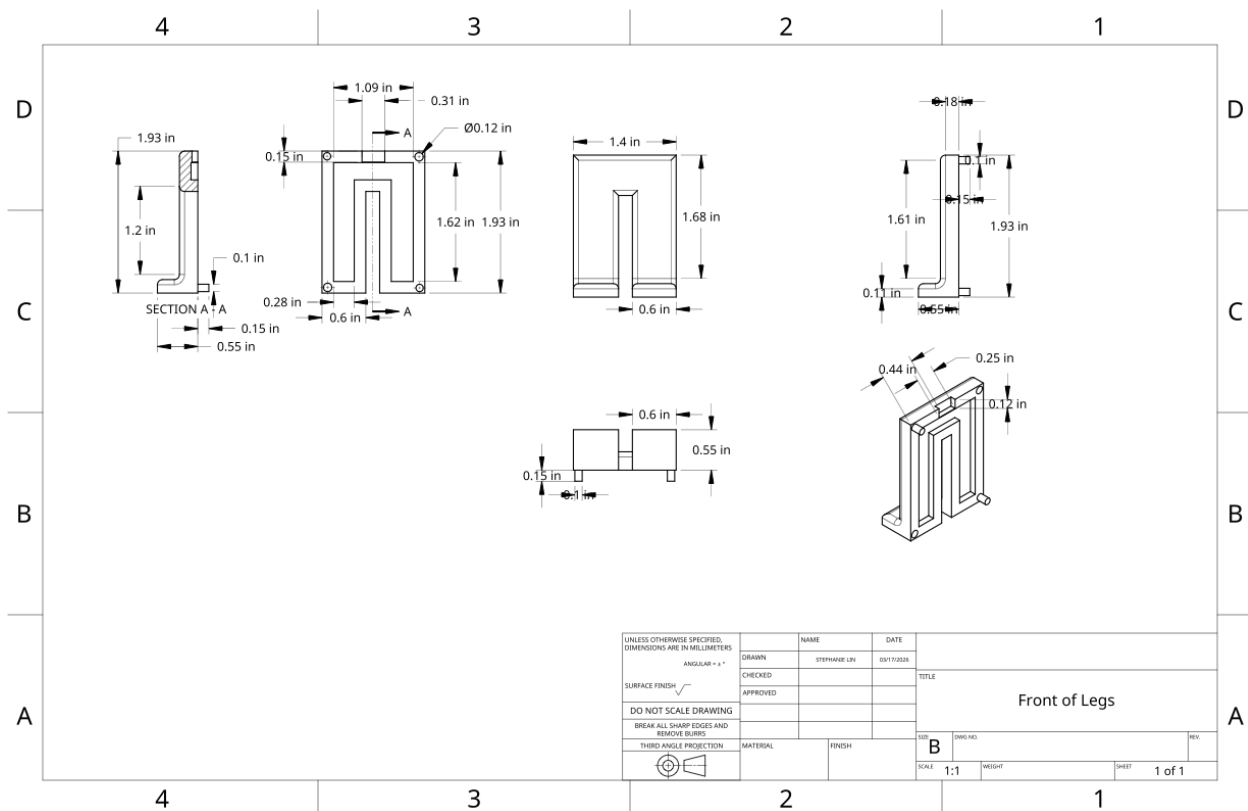


Figure 3.1. Front of Legs Part Drawing

The CTF dimensions for the back of the legs are made to allow for the control alignment with the torso, allowing for stability of the figure, and the structural integrity of the leg assembly. The 1.93 inch overall leg length and 1.79 inch in reference height are CTF because they set the vertical position of the legs relative to the torso and align with the front-leg geometry to ensure the feet sit flat and the figure stands. The 0.61 inch spacing between the mounting region and the lower features determines the effectiveness of the mating torso sockets and directly impacts the assembly and poseability. The Ø0.12 in pin features and the 0.15-0.31 inch associated widths are also CTF since they define the conditions for the connection between the two halves of the legs. During the prototyping design process, excessive variation from these dimensions led to loose

joints and prevented the connection between the two halves. The 0.10-0.18 in local thickness and the R0.07 inch fillet radius are CTF dimensions for durability and manufacturability. They control stress concentration at corners and ensure sufficient wall thickness to avoid breakage while still allowing the parts to be injection-moldable. The 0.46 inch lateral dimension of the leg part is CTF because it prevents interference with adjacent parts while maintaining desired visual proportions.

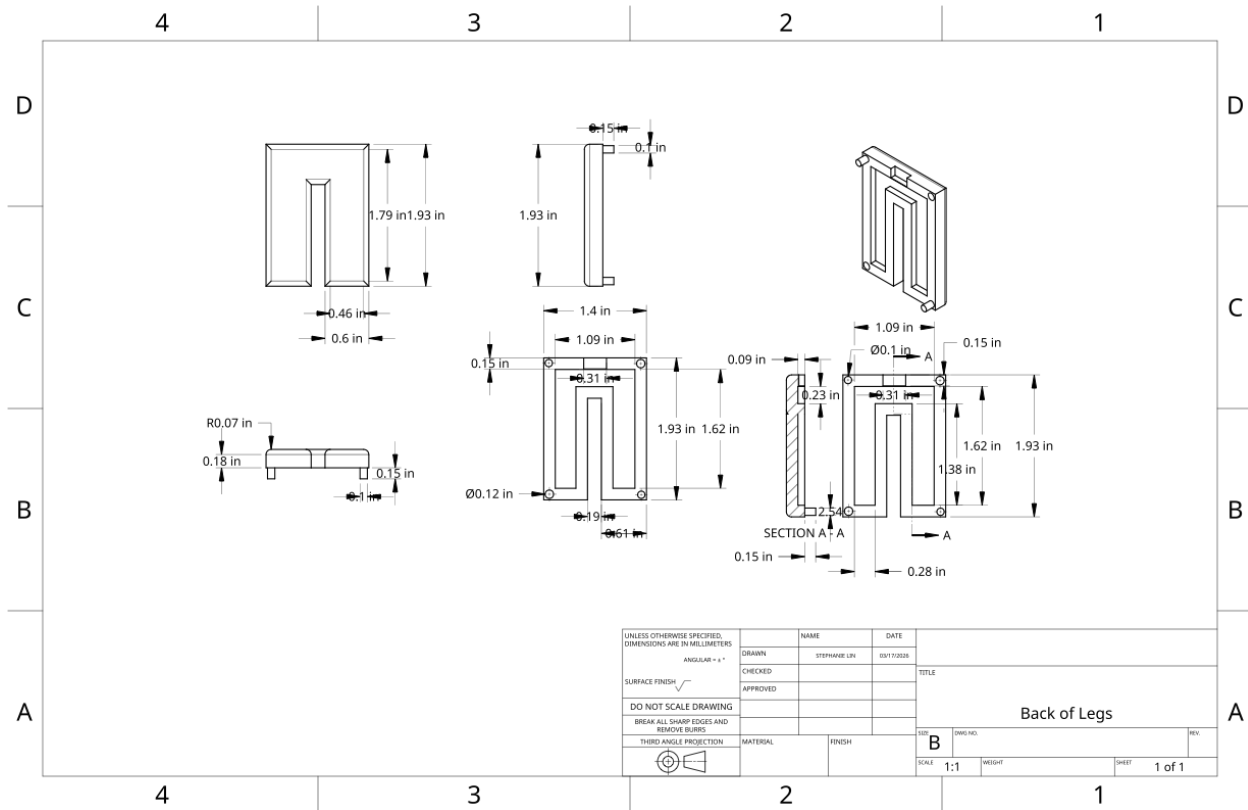


Figure 3.2. Back of Legs Part Drawing

Table 3.1. CTF Dimensions and Tolerance of Legs

Dimension Description	Nominal (in.)	Tolerance ($\pm$ in.)	Min (in.)	Max (in.)
Overall Leg Height	1.93	0.01	1.92	1.94
Upper Leg Width	0.55	0.01	0.54	0.56
Lower Leg Width	0.44	0.01	0.43	0.45
Inner Slot Width	0.25	0.01	0.24	0.26
Thin Wall Thickness	0.10	0.01	0.09	0.11
Thick Wall Thickness	0.18	0.01	0.17	0.19

Table 3.2 Pin Hole Interface Worst Case Clearance for Legs

Case	Calculation	Clearance (in)
Minimum Clearance	0.119-0.122	-0.003
Maximum Clearance	0.121-0.119	0.003

The clearance range for the leg pivot pin-hole interface is between -0.003 in. and 0.003 in. The negative clearance indicates that interference may occur under worst-case conditions if both the leg pivot pins and the torso pivot holes are manufactured at their extreme tolerance limits. This indicates that the current combinations of a light interference fit and a small clearance fit, meaning clearance is not guaranteed in all cases.

However, under typical injection-molding conditions, it is unlikely that both components will simultaneously be manufactured at their extreme tolerance limits. The maximum clearance of 0.003 should still allow the leg to pivot with very minimal looseness while still maintaining alignment at the connecting joint.

Mounting Slot Interface

A key functional feature of the toy is the intentional “breakability” of the torso and leg sections; therefore, clearances between these components must be carefully controlled to ensure proper separation and reassembly.

Table 4.1 Mounting Slot Interface Worst Case Clearance

Case	Calculation	Clearance (in.)
Length Minimum Clearance	0.299-0.26	0.039
Length Maximum Clearance	0.301-0.24	0.061
Width Minimum Clearance	0.79-0.11	-0.031
Width Maximum Clearance	0.081-0.09	-0.009
Depth Minimum Clearance	0.149-0.11	-0.039
Depth Maximum Clearance	0.151-0.09	0.061

Design Documentation for Catapult Base

The CTF dimensions for the castle catapult base are primarily associated with the pivot interface that connects the catapult arm to the base. The most critical feature is the pivot slot at the top of the base, which controls the fit and rotational motion of the arm. The slot width was designed to provide a clearance fit with the catapult arm, allowing smooth rotation without interference. A tolerance of ± 0.02 inches was applied to the slot width to ensure ease of assembly under typical injection molding. A tighter tolerance of ± 0.005 inches was applied to the pivot contact region to maintain proper alignment and minimize wobble during operation. Additionally, a tolerance of ± 0.02 inches was applied to the pivot slot length, as this feature is less critical to motion and primarily affects positioning. These tolerances were selected to balance manufacturability with functional performance, ensuring reliable operation while remaining achievable within standard injection molding capabilities.

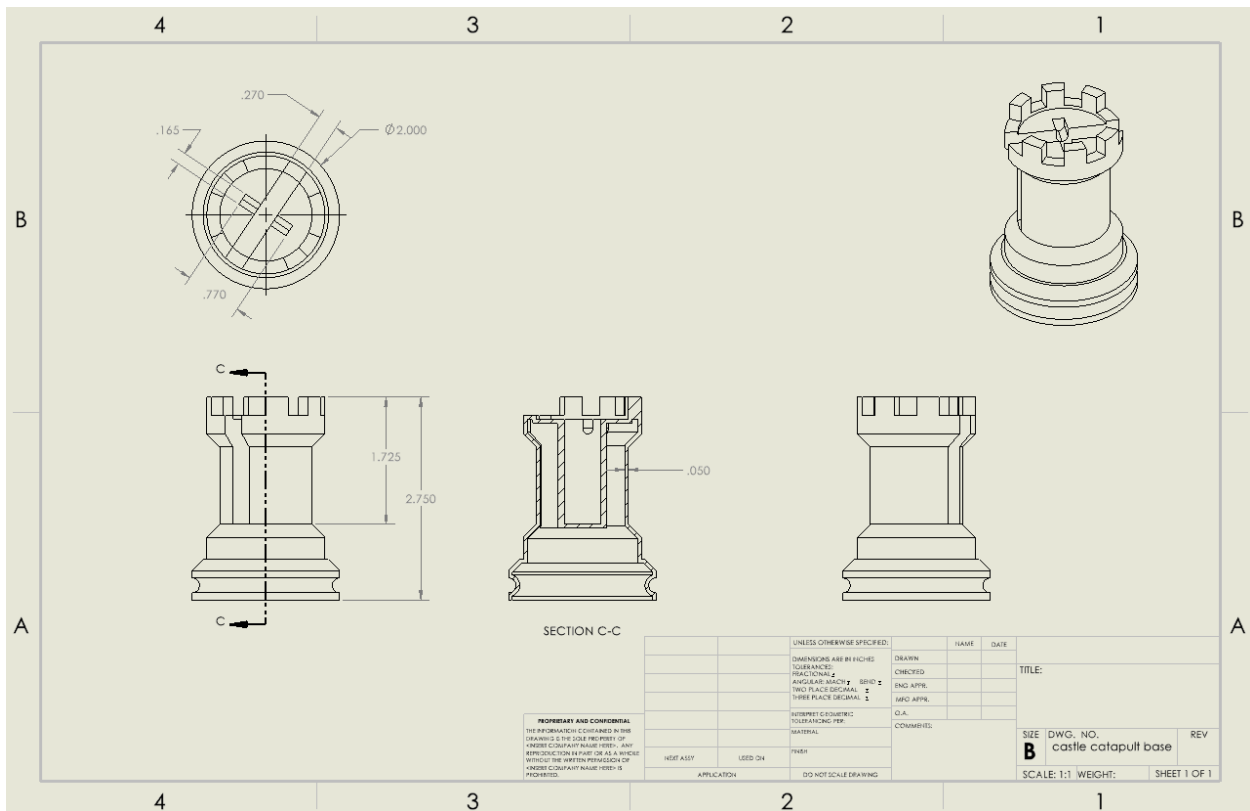


Figure 5.1. Castle Catapult Base Part Drawing (with cross-section view)

Table 5.1 CTF Dimensions and Tolerances for Catapult Base

Dimension Description	Nominal (in.)	Tolerance ($\pm$ in.)	Min (in.)	Max (in.)
Catapult Arm Slot Width	0.27	0.02	0.25	0.29
Catapult Pivot Slot Length	0.75	0.02	0.73	0.77
Catapult Pivot Slot Width	0.16	0.005	0.155	0.165

The pivot slot width is the primary CTF dimension, as it directly controls the fit and rotational motion of the catapult arm. A worst-case tolerance analysis was performed to evaluate the clearance between the pivot slot and the catapult arm, as shown in Table 5.2.

Table 5.2 Catapult Base Pivot Interface Worst Case Clearance

Case	Calculation	Clearance (in.)
Minimum Clearance	$0.155 - 0.16$	-0.005
Maximum Clearance	$0.165 - 0.14$	0.025

The clearance range is between -0.005 inches and 0.025 inches. The negative clearance indicates that interference may occur under worst-case conditions if both components are at their extreme tolerance limits. This suggests that the current tolerance range may be slightly too large to guarantee a clearance fit in all cases. However, under typical injection molding conditions, parts are unlikely to reach worst-case limits simultaneously, and the design is expected to function properly in most cases. The maximum clearance of 0.025 inches ensures that the catapult arm can rotate freely without interference.

A tighter tolerance (± 0.005 inches) was applied to the pivot slot width because this feature directly affects motion and alignment. In contrast, larger tolerances (± 0.02 inches) were used for non-critical dimensions such as slot width and slot length to simplify manufacturing and reduce production cost. These tolerance selections balance manufacturability with functional performance, ensuring smooth rotation, acceptable alignment, and ease of assembly. To fully eliminate potential interference, a slight increase in slot width or reduction in tolerance could be implemented.

Design Documentation for Catapult Arm

The catapult arm is a motion-critical component whose function is to rotate within the catapult base pivot slot while withstanding repeated loading during launching. Thus, the CTF dimensions for the catapult arm control rotational freedom, structural strength, and overall user experience.

The most important CTF feature is the catapult pivot shaft width, which fits into the pivot slot in the catapult base. This is a CTF dimension because it determines whether the arm will rotate freely, harshly due to friction, or in a wobbly manner. This dimension was designed to provide a nominal clearance fit with the base pivot slot width defined in Table 5.1. The catapult arm thickness is also a CTF dimension because it determines if the arm can move freely when rotated.

The overall arm length is also a CTF dimension because it determines the launch angle, affecting the projectile travel distance and user experience. Additionally, the projectile cup is considered CTF because it determines how the projectile sits and releases during launch. The cup diameter and depth ensure that the projectile remains in the cup during arm rest and acceleration but exits at the end of arm travel. Imprecise tolerances could result in early release or failure to launch.

Table 6.1 CTF Dimensions and Tolerances for Catapult Arm

Dimension Description	Nominal (in.)	Tolerance ($\pm$ in.)	Min (in.)	Max (in.)
Catapult Arm Width	0.25	0.01	0.24	0.26
Catapult Arm Length	2.50	0.03	2.47	2.53
Projectile Cup (outer diameter)	1.50	0.02	1.48	1.52
Catapult Pivot Shaft Width	0.15	0.01	0.14	0.16

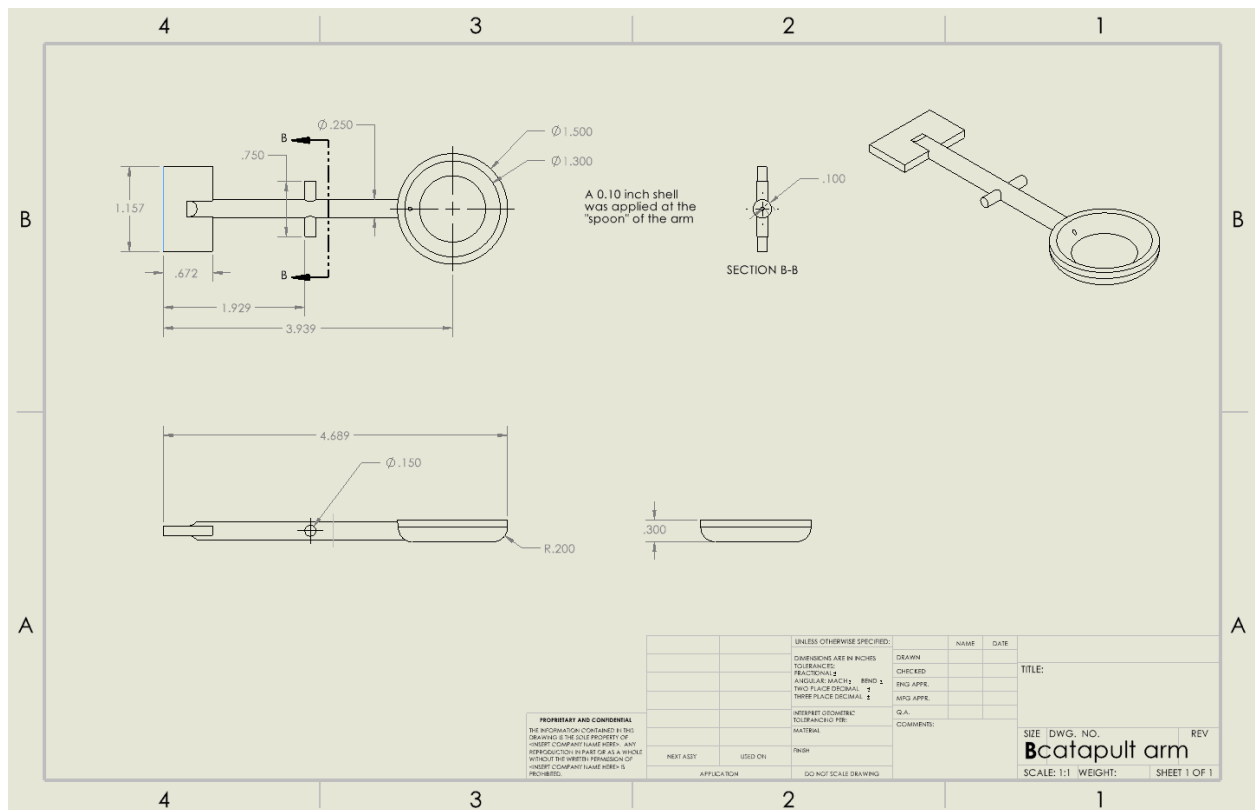


Figure 6.1 Catapult Arm Part Drawing (with cross-section view)

3D Exploded View of Toy Assemblies

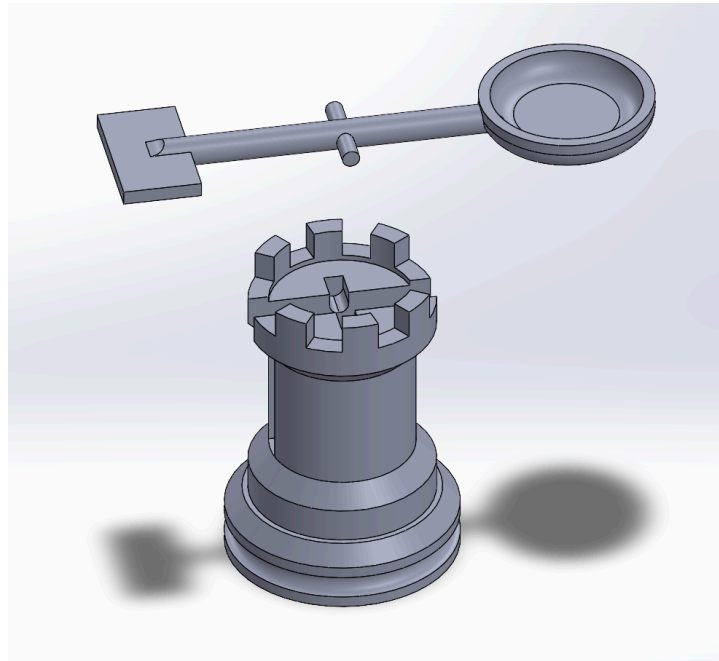


Figure 7.1 Catapult Exploded View Assembly

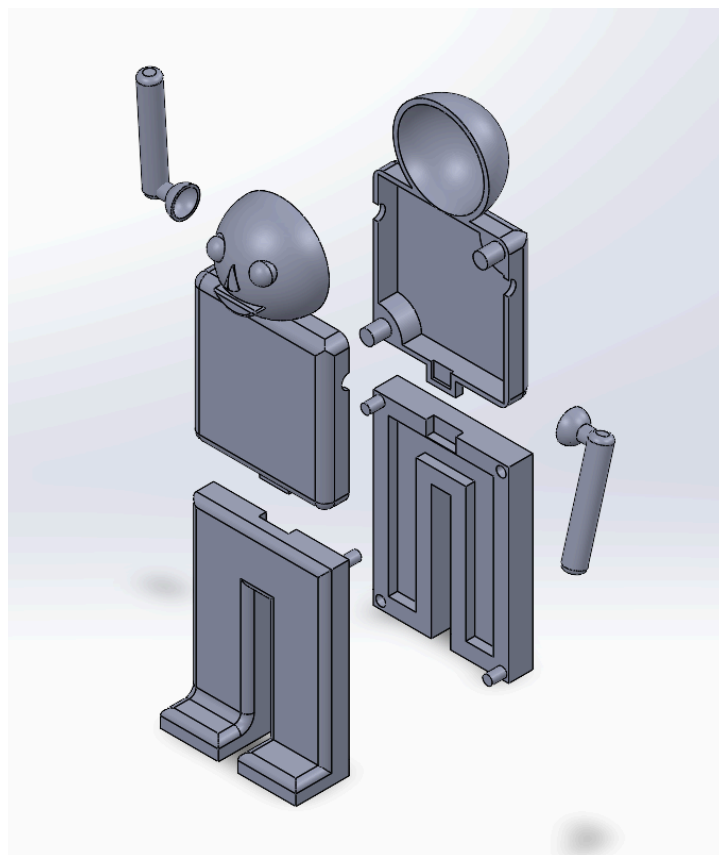


Figure 7.2 “Barbarian” Target Break-Apart Figurine Exploded View

Design for Manufacturability

To maximize cost efficiency, every component was shelled in order to lower the use of total plastic. To allow for the lower and upper bodies to be injection moldable, both individual parts would be split into two, simplifying the mold design process during manufacturing. One half would have cylindrical pins that would fit perfectly into the holes of the other half.

Although the arms of the upper body do not comply with standardized choking standards, it was designed with the idea that during manufacturing, the two halves of both the lower and upper bodies would be glued, making it impossible for the arms to be separated from the upper body, thus, not a choking hazard.

Manufacturing Plan

The toy components are designed to be manufactured using injection molding, the most common and cost-effective process for producing plastic toys at high volumes. The plastic parts estimator shown in Table 1 was used to approximate manufacturing costs for each component of the toy.

units >	Projected Area		Cavities per Tool		Injection Pressure		Press Size		Press Cost		Cycle Time		Yield Rate		Parts / Hour		Molding Cost		Material		Specific Gravity		Part Volume		Part Volume		Part Weight		Raw Material Cost		Part Material Cost		Total Part Cost †		Tools			Run time/day		Working days/week		Parts / year		Total Cost		Part Weight / Year		Scrap Parts		Scrap Cost	
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Part Name																																																			
1 BlankTorso	2.9	18	6	310	No	42	30	99.999	2160	\$0.019	PP - cp	0.91	3.93	0.24	0.008	\$0.68	\$0.005	\$0.02	1	23.5	7	18,476,455	\$458,399	145,781	184,766	\$4,584.04																									
2 FaceTorso	2.9	18	6	313	No	42	30	99.999	2160	\$0.019	PP - cp	0.91	3.77	0.23	0.008	\$0.68	\$0.005	\$0.02	1	23.5	7	18,476,455	\$454,269																												
3 BaseCat	3.1	18	6	339	No	42	30	99.999	2160	\$0.019	PP - cp	0.91	15.24	0.93	0.031	\$0.68	\$0.021	\$0.04	1	23.5	7	18,476,455	\$743,402																												
4 LeverCat	3.3	18	6	351	No	42	30	99.999	2160	\$0.019	PP - cp	0.91	8.03	0.49	0.016	\$0.68	\$0.011	\$0.03	1	23.5	7	18,476,455	\$561,661																												
5 Legs1	3.1	18	6	335	No	42	30	99.999	2160	\$0.019	PP - cp	0.91	9.28	0.57	0.019	\$0.68	\$0.013	\$0.03	1	23.5	7	18,476,455	\$593,053																												
6 leg2	3.1	18	6	335	No	42	30	99.999	2160	\$0.019	PP - cp	0.91	9.28	0.57	0.019	\$0.68	\$0.013	\$0.03	1	23.5	7	18,476,455	\$593,053																												
Total:																									\$0.18			\$3,403,837																							

Table 8.1 Plastic Parts Estimator

All component materials were specified as PP-cp (polypropylene copolymer). PP-cp was selected because it provides a good balance of impact resistance, flexibility, and durability, which are important for toys subjected to repeated forces, such as when the barbarian target falls or when the catapult arm strikes a surface. PP-cp also has a relatively low material cost and good moldability, making it well-suited for mass production. Additionally, PP-cp is a recyclable material, helping to reduce environmental impact.

Following injection molding, the torso and leg components would be fixed together. Each component includes two pegs and matching holes with appropriate tolerances to facilitate assembly, which would be achieved using either heat staking or adhesive bonding. Heat staking involves melting the plastic pegs and bonding the parts together instead of using adhesive to glue them. This method would most likely be used in manufacturing as it is cost-effective and quick, adding about \$0.01 cost per unit. Alternatively, cyanoacrylate adhesive (super glue) could be used, but may cost up to \$0.05 per unit.

Finally, facial features or other graphical details must be applied to the barbarian and the catapult toy. For the purpose of mass-produced plastic toys, pad printing is the industry standard. If three distinct colors are applied to the toy, the expected additional cost will be about \$0.05 per unit.

After considering the additional costs of post-injection molding processes, the total estimated cost of manufacturing the Happy Meal toy is \$0.24 per unit. This price is negligible and does not significantly affect the profit margin of the Happy Meal, providing an engaging interactive experience to the customer with little impact to overall production costs. The manufacturing processes selected also allow for high production rates, which is critical for the large-scale demand required for a Happy Meal toy.

Sustainability

The team aimed to find ways to implement the existing food packaging into the toy's design so that there is no plastic projectile included in the toy. Instead, the user is encouraged to repurpose food wrappers or napkins that already come with the meal as projectiles, reducing material waste and promoting sustainability. In the team's advertisement, users are also encouraged to use the catapult to launch the repurposed projectile into the trash to promote proper disposal of waste and reduce littering. The toy will also be produced with recyclable PP-cp plastic, helping to reduce environmental impact when the toy is properly disposed of.

Marketing Analysis

The target demographic for this Happy Meal toy is children between the ages of approximately 4 and 9 years old. Children in this age range are typically attracted to toys that involve physical interaction and competition. The medieval theme provides a simple story context in which children can imagine defending a castle or defeating an enemy character. The catapult launching mechanic adds a skill-based challenge that encourages repeated play.

The toy also incorporates several parent-friendly features. The design does not come with a small plastic projectile, reducing the risk of lost pieces, bodily harm to others, or plastic waste. All components are large enough to meet choke hazard safety requirements, and the durable material used ensures that the toy can be reused multiple times.



Figure 8.1 Siege the Waste™ Marketing Sheet

Conclusions

The Siege the Waste™ toy successfully meets the design objective of being creative, safe, sustainable, and manufacturable. The catapult launching mechanism and break-apart target create an engaging and interactive toy experience for children. The design also incorporates sustainability by encouraging the reuse of packaging materials as projectiles instead of including an additional plastic component. This promotes sustainability ideals within children while simultaneously reducing material waste. Since the toy uses simple injection-molded components and snap-fit assembly, it can be produced efficiently at high volumes while maintaining low manufacturing costs. As a result, the toy represents a practical and innovative concept for a McDonald's Happy Meal toy.

Appendix A. Duke Toy Fair Deliverables and Booth Setup



Figure A1. Siege the Waste™ Toy Fair Booth Setup

Figure A2. Siege the Waste™ Video Advertisement
[Video Link](#)

Team Reflection

What worked well: Highlight strengths in your team's collaboration, processes, and outcomes.

Overall, the team dynamic was very positive throughout the project. All team members communicated respectfully and contributed to the development of the toy concept and design documentation. Responsibilities were divided effectively, and each team member stepped up when they felt they weren't contributing enough. For example, Guido volunteered to create the advertisement for the toy fair showcase while the other team members worked on the deliverables required for the showcase. Despite limited opportunities to meet in person, the team was able to collaborate successfully through asynchronous work and shared progress updates. This flexibility allowed the project to continue moving forward even with scheduling constraints.

Challenges faced: Discuss any difficulties encountered and how they were addressed.

One of the main challenges the team faced was difficulty coordinating meeting times due to conflicting schedules. As a result, much of the work was completed asynchronously, which occasionally led to delays in decision-making and coordination. Additionally, the team did not initially standardize the CAD software used for design. Team members used both Onshape and SolidWorks, which created compatibility issues when assembling the final design. This required one team member to recreate all parts in SolidWorks in order to create the complete assembly and exploded views.

Lessons learned: Reflect on key takeaways from the workshop and how these can be applied in future projects.

A key lesson learned from this project is the importance of early planning and coordination. Establishing a consistent schedule for meetings would improve communication and reduce last-minute work. The team also recognized the importance of agreeing on a common CAD application at the start of the project to avoid compatibility issues and repeated work. Additionally, starting work earlier and distributing tasks more evenly over time would reduce time pressure leading up to the toy fair showcase.

Growth and improvements: Evaluate how your team improved over the course of the project and what you might do differently next time.

Over the course of the project, the team improved its ability to collaborate effectively despite scheduling challenges. Team members adapted by communicating more clearly through asynchronous methods and taking initiative in completing assigned tasks. For future projects, the team would aim to establish clearer expectations at the beginning, including scheduling regular meetings, standardizing tools such as CAD software, and setting soft deadlines. These improvements would help increase efficiency, reduce last-minute work, and enhance overall project quality.